

INTERNATIONAL STANDARD

**Optical amplifiers –
Part 5-2: Qualification specifications – Reliability qualification for optical fibre
amplifiers**



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**Optical amplifiers –
Part 5-2: Qualification specifications – Reliability qualification for optical fibre
amplifiers**

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INTERNATIONAL ELECTROTECHNICAL COMMISSION

OPTICAL AMPLIFIERS –**Part 5-2: Qualification specifications – Reliability qualification
for optical fibre amplifiers**

FOREWORD

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International Standard IEC 61291-5-2 has been prepared by subcommittee 86C: Fibre optic systems and active devices, of IEC technical committee 86: Fibre optics.

This second edition cancels and replaces the first edition published in 2002. It constitutes a technical revision.

This edition includes the following significant technical changes with respect to the previous edition:

- a) removal of the contents on the relating quality management system from scope, terms and definitions, and the reliability requirements;
- b) moving fit-rate calculation to Annex B (informative);
- c) change of requirements for shock test;
- d) amendment of abbreviations related to changes a) and b).

The text of this International Standard is based on the following documents:

CDV	Report on voting
86C/1376/CDV	86C/1426/RVC

Full information on the voting for the approval of this International Standard can be found in the report on voting indicated in the above table.

This document has been drafted in accordance with the ISO/IEC Directives, Part 2.

A list of all parts in the IEC 61291 series, published under the general title *Optical amplifiers*, can be found on the IEC website.

The committee has decided that the contents of this document will remain unchanged until the stability date indicated on the IEC website under "<http://webstore.iec.ch>" in the data related to the specific document. At this date, the document will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

OPTICAL AMPLIFIERS –

Part 5-2: Qualification specifications – Reliability qualification for optical fibre amplifiers

1 Scope

This part of IEC 61291 applies to optical amplifiers (OAs) and optically amplified, elementary sub-systems for terrestrial applications, using active fibres (optical fibre amplifiers (OFAs)) containing rare-earth dopants, which are commercially available.

The black box approach is used in this document. The black box approach is adopted in order to give product specifications which are independent of OA implementation details. For reliability qualification purposes, some information about the internal components is needed; these internal parts are themselves treated as black boxes. This document gives requirements for the evaluation of OA reliability by combining the reliability of such internal black boxes.

The object of this document is to specify the minimum list of reliability qualification tests, requirements on failure criteria during testing and on reliability predictions, and give the relevant normative references to establish a standard method for the assessment of the reliability of OFA devices and sub-systems in order to minimize risks and to promote product development and reliability qualification.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60050-731, *International Electrotechnical Vocabulary – Chapter 731: Optical fibre communication*

IEC 60068-2-2:2007, *Environmental testing – Part 2-2: Tests – Test B: Dry heat*

IEC 60068-2-14, *Environmental testing – Part 2-14: Tests – Test N: Change of temperature*

IEC 60068-2-21, *Environmental testing – Part 2-21: Tests – Test U: Robustness of terminations and integral mounting devices*

IEC 60068-2-27, *Environmental testing – Part 2-27: Tests – Test Ea and guidance: Shock*

IEC 60068-2-31, *Tests – Test Ec: Rough handling shocks, primarily for equipment-type specimens*

IEC 60068-2-78, *Environmental testing – Part 2-78: Tests – Test Cab: Damp heat, steady state*

IEC 61291-1, *Optical fibre amplifiers – Part 1: Generic specification*

IEC 61300-2-4, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 2-4: Tests – Fibre/cable retention*

IEC 62005-9-1, *Fibre optic interconnecting devices and passive components – Reliability – Part 9-1: Qualification of passive optical components*

IEC 62005-9-2, *Reliability of fibre optic interconnecting devices and passive optical components – Part 9-2: Reliability qualification for single fibre optic connector sets – Single mode*

TIA 455-11, *FOTP-11 Vibration Test Procedure for Fiber Optic Components and Cables*

3 Terms, definitions and abbreviated terms

3.1 Terms and definitions

For the purposes of this document, the terms and definitions given in IEC 61291-1, IEC 60050-731 and the following apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at <http://www.electropedia.org/>
- ISO Online browsing platform: available at <http://www.iso.org/obp>

3.1.1 failure

non-compliance to product specification or change in parameters as agreed by the customer and supplier

3.1.2 OFA manufacturer OFAM

manufacturer who provides optical fibre amplifier (OFA) devices or subsystems meeting the requirements of the applicable product specification (PS)

Note 1 to entry PS includes the reliability requirement.

3.2 Abbreviated terms

EDFF	erbium doped fluoride fibre
EDSFA	erbium doped silica fibre amplifier
EDTF	erbium doped tellurite fibre
FIT	failure in time
FFS	for further study
OA	optical amplifier
OFA	optical fibre amplifier
OFAM	optical amplifier manufacturer
OFAM	optical fibre amplifier manufacturer
PDFF	praseodymium doped fluoride fibre
PS	product specification
RH	relative humidity
DFF	thulium doped fluoride fibre
UCL	upper confidence level

4 Reliability requirements

4.1 Tests

4.1.1 General

An optical fibre amplifier (OFA) device or sub-system is an assembly of various parts and components of different nature. A main point in the reliability testing of the OFA device or sub-system is to ensure the reliability of each part and basic manufacturing process used and to provide a route that may result in an understanding of design margins. For the purpose of this document, each internal part shall be seen as a black box.

This document is based on the assumption that the reliability of an optical amplifier (OA) can be evaluated with sufficient confidence from the failure in time (FIT) rates of its internal black boxes when the assembly process of the constituents has been qualified.

The procedures to qualify the assembly process are described in 4.1.3.

The reliability assurance is confirmed by the test procedure described in 4.1.4 carried out on the OFA device or sub-system as a whole.

The basic parts usually constituting an OFA are listed below:

- passive optical components;
- opto-electronic components;
- doped fibres or doped fibre modules;
- optical connectors;
- electronics;
- others (to be specified);
- mechanical packaging.

The OFA manufacturer (OFAM) should declare the number and type of the internal black boxes constituting the OFA and give the failure rates (in FIT) for each of them.

The OFA failure rate should be calculated by suitably combining the FIT rates of its internal parts, as described in Annex B.

4.1.2 Reliability qualification of components

Table 1 and Table 2 give the minimum list of tests to be performed on the various parts, where used, constituting the OFA in order to guarantee the claimed reliability level. Normative references for tests and test conditions are given in Annex A.

Table 1 – Minimum test list for passive optical components, pump laser modules, monitor diode modules and optical connectors

Component	Reference
Pump laser diode	IEC 62572-3 (informative)
Photo diode	To be defined
Passive optical components	IEC 62005-9-1
Optical connectors	IEC 62005-9-2
Variable optical attenuators	To be defined

Table 2 – Minimum test list for doped fibre

Test
Proof test
Hydrogen aging
Fibre coating strip force (for non-hermetic fibre only)

A set of test conditions suitable to the test lists in Table 1 and Table 2 to assess the reliability of OFA components (seen as black boxes) is given for reference in Annex A. These test conditions specify the common practice in the OFA manufacturing industry.

4.1.3 Reliability qualification of the OFA assembly process

Fibre arrangement and assembly of the optical and electrical components are very important aspects for assessing the reliability of an OFA. In particular, the fibre winding and splice process is one of the most critical steps in the OFA assembly process.

The splice process shall be qualified according to the relevant fibre and coating material.

Table 3 indicates the tests required on splices.

Table 3 – Tests required for splices

Tests	Conditions	Reference
High temperature storage	+85 °C 2 000 h	IEC 60068-2-2, Tests B
Change of temperature (Thermal cycling) ^a	-40 °C / +85 °C 2 °C/min <i>Q</i> = 100 cycles <i>I</i> = 500 cycles	IEC 60068-2-14
Damp heat	40 °C 93 % RH 500 h	IEC 60068-2-78
Shock	1 000 m/s ² 6 ms 6 times/axis	IEC 60068-2-27
Vibrations	5 Hz to 50 Hz, 15 m/s ² 50 Hz to 500 Hz, 30 m/s ² 3 axis Duration 15 sweeps	TIA 455-11
Robustness test (Pull test)	5 N 10 s	IEC 60068-2-21
^a <i>Q</i> : data for qualification, <i>I</i> : data for information.		

The number of samples is to be agreed between customer and supplier according to the level of confidence and the level of reliability required.

4.1.4 Reliability qualification of the OFA device or sub-system

A reliability qualification procedure related to the complete OFA device or sub-system is described in Table 4. It gives the minimum list of tests to be performed on OFA devices and sub-systems in order to assure reliability. Normative references, where tests and test conditions are specified, are given in Annex A.

The purpose of the testing is to assess the prediction of the failure rate of the complete OFA performed according to the procedure of 4.1.2.

On the basis of the reliability assurance required for the reliability tests for the OFA internal black boxes, the sampling level is generally low (for example a few samples for each amplifier type).

In some specific cases (for example, non-silica glass OFAs), the use of adhesives in the OFA can be considered as a critical process and shall be qualified separately. Depending on the possible function of the adhesive (mechanical anchoring, splice protection, index matching, etc.), the different failure modes shall be addressed and supported by reliability data.

Table 4 – Minimum list for tests required on OFA devices and sub-systems

Test		Condition	Duration	Operating	Samples
Temperature storage		$T_{\text{stg,min}}/T_{\text{stg,max}}$	72 h	-	3
Temperature cycling ^a		$T_{\text{stg,min}}/T_{\text{stg,max}}$ Ramp > 1 °C/min	$Q = 100$ cycles $I = 500$ cycles	Y	3
Damp heat		85 °C to 85 % RH ^b	$Q = 500$ h $I = 1\,000$ h	-	3
Mechanical test	Shock (≤ 0,225 Kg) ^c	3 000 m/s ² 3 ms pulse	5 shocks per direction, 6 directions	-	3
		See table below ^d	See table below ^c	-	
	Shock (> 0,225 Kg) ^c	500 m/s ² , 11 ms pulse	5 shocks per direction, 6 directions	-	3
		See table below ^d	See table below ^c	-	
	Shock: circuit pack, blades and racks ^c	See table below ^d	See table below ^c	-	3
	Vibration (≤ 1 Kg)	200 m/s ² , 20 Hz to 2 000 Hz to 20 Hz 3-axis, 4 cycles per axis, 2 h/axis	4 mins/cycle	-	3
	Vibration (> 1 Kg)	5 Hz to 50 Hz, 15 m/s ² 50 Hz to 500 Hz, 30 m/s ² 3-axis	0,1 oct/min 0,25 oct/min	-	3
	Vibration: circuit pack, blades and racks	5 Hz to 100 Hz to 5 Hz, 10 m/s ² 3-axis	0,25 oct/min	-	3
Pull ^e		5 N, 10 N and 100 N ^d		-	≥ 12
Endurance ^a		$T_{\text{op,max}}/P_{\text{nom}}$	$Q = 1\,000$ h $I = 2\,000$ h	Y	3

Test	Condition	Duration	Operating	Samples								
No failures are allowed												
Tests may be performed sequentially or in parallel.												
If the storage temperature is lower than the specified humidity temperature, another test at $T_{\text{stg,max}}$ and RH $\geq$ 85 % is done.												
If the maximum storage temperature is similar to the maximum operating temperature, the storage test at $T_{\text{stg,max}}$ is not required and is replaced by the endurance test. The endurance test is performed at fixed pump current, output power or pump monitor current (nominal values given in the product specification (PS)). To check any drift of the amplifier, relevant parameters [such as output power, pump parameters (if accessible) etc.] may be measured during the test.												
A reference to the temperature cycle test method is provided in Annex B.												
$T_{\text{stg,min}}$ is the OFA minimum storage temperature;												
$T_{\text{stg,max}}$ is the OFA maximum storage temperature;												
$T_{\text{op,max}}$ is the OFA maximum operating temperature;												
P_{nom} is the OFA nominal output power.												
^a Q qualification, I : information.												
^b Damp heat: the damp heat test at 85 °C/85 % RH has been advocated by some manufacturers, as justified by observations of conditions within peculiar environments. These test conditions may be used for OFAs to be deployed in very hard environments. Otherwise, the damp heat test at 40 °C/93 % RH may be used.												
^c Either controlled shock test or drop test can be applied.												
^d Mechanical test: shock (free drop, IEC 60068-2-27 and IEC 60068-2-31)												
<table><tr><th>Mass</th><th>Drop height</th></tr><tr><td>kg</td><td>mm</td></tr><tr><td>$0 \leq \text{mass} < 10$</td><td>100</td></tr><tr><td>$10 \leq \text{mass} < 25$</td><td>75</td></tr></table>					Mass	Drop height	kg	mm	$0 \leq \text{mass} < 10$	100	$10 \leq \text{mass} < 25$	75
Mass	Drop height											
kg	mm											
$0 \leq \text{mass} < 10$	100											
$10 \leq \text{mass} < 25$	75											
^e Pigtail testing – pull test (the first figure in each row below is the outer diameter of the buffered or cabled fibre to which the specified test conditions do apply)												
<table><tr><td rowspan="3">Cable retention (pull)</td><td>reinforced fibre: 20 N to 100 N, 3 times, 5 s pulls</td><td rowspan="3">IEC 61300-2-4</td></tr><tr><td>secondary coated fibre: 10 N, 3 times, 5 s pulls</td></tr><tr><td>primary coated fibre: 5 N, 3 times, 5 s pulls</td></tr></table>					Cable retention (pull)	reinforced fibre: 20 N to 100 N, 3 times, 5 s pulls	IEC 61300-2-4	secondary coated fibre: 10 N, 3 times, 5 s pulls	primary coated fibre: 5 N, 3 times, 5 s pulls			
Cable retention (pull)	reinforced fibre: 20 N to 100 N, 3 times, 5 s pulls	IEC 61300-2-4										
	secondary coated fibre: 10 N, 3 times, 5 s pulls											
	primary coated fibre: 5 N, 3 times, 5 s pulls											

It is essential that the evaluated OFA devices or sub-systems are entirely representative of standard production and have passed all the production procedures and/or specified (where applicable in the DS) burn-in and screening procedures.

Aspects of the test conditions not provided in this document are given in the relevant PS.

4.1.5 Structural similarity

Where a range of optical amplifier modules is produced by an OFAM, there can be some significant structural similarity between different type codes. A combination of results from different test programmes, where appropriate, is therefore permitted. Although an understanding of similarity is necessary, it is also important to provide an assessment of differences. All assessments should be compared on a test-by-test basis.

Consideration should be given to the fact that minor differences in technology or processing can have a major impact on reliability, whilst not being apparent during quality assessment.

Evidence should be presented which demonstrates that all results are directly relevant.

Annex A (normative)

Procedures for reliability testing of OFA components

A.1 General

This normative annex contains test lists and test conditions suitable to assess the reliability of OFA components (seen as black boxes). These lists reflect the common practice in the OFA manufacturing industry and are given as a reference, together with the relevant published IEC documents, for OFA customers.

NOTE IEC references included in Clause A.2 are subject to updating to include new documents as they are published.

A.2 Tests required for passive optical components

IEC 62005-9-1 describes the tests required for passive optical components used in the OFA device or sub-system in controlled environment and in outside plant environment.

Refer to IEC 62005-9-2 for definition of the tests required on optical connectors.

A.3 Tests required for the doped fibre

Table A.1 indicates the tests required for the doped fibre.

Table A.1 – Tests required for the doped fibre

Test	Condition
Proof test	Proof test based on deployment condition
Hydrogen aging	See note below
Fibre strip force (for non-hermetic fibre only)	Mechanical stripping

Hydrogen aging: Regardless of hydrogen protection, the OFA maker shall be able to demonstrate that the OFA will meet this requirement in the presence of as much as 10⁻² atmosphere of hydrogen over the life of the device. This should be done by exposing the fibre to accelerating temperatures and pressures. The proper choice of accelerating temperature and pressure varies by fibre design and method of manufacture. The applied test method shall be agreed upon between the OFAM and user.

The bending tests on the doped fibre are performed with bending radii similar to those encountered in the conditions of use; the test conditions shall be specified (bare fibre, fibre tested on a bobbin or mandrel, etc.). The tests are performed preferentially on the fibre as supplied; if needed, additional tests on the fibre in a sub-assembly can be performed. The test method on the fibre in a sub-assembly is for further study (FFS).

Temperature and humidity impacts are currently under study by the suppliers of doped fibre.

Here, the doped fibre is intended to be a silica-based fibre such as that used in erbium doped silica fibre amplifier (EDSFAs). On the other hand, non-silica-based fibres comprising, for example, erbium doped fluoride fibre (EDFFs), erbium doped tellurite fibre (EDTFs), praseodymium doped fluoride fibre (PDFFs), and thulium doped fluoride fibre (TDFFs), usually need to be housed in a sealed package to avoid moisture damage. Therefore, the sealed package is treated as a doped fibre module and qualified separately taking into account the peculiarity of its mechanical properties.

Annex B (informative)

Reliability calculations

B.1 Reliability calculation

A reliability prediction regarding the complete OFA device or sub-system is provided by the OFAM, based on the failure rates [in FIT (failure in time)] of the internal black boxes composing the OFA.

The failure rates of the internal black boxes shall be given by the OFAM taking into account the basic values issued from the cumulated component-hours issued from the different components included in the amplifier. The calculations for each internal black box shall be based on the current standards regarding reliability calculations.

The reliability calculations will also include the wear-out failures, which mainly concern the pump laser module for the optical fibre amplifier. The FIT figures given for each internal black box shall take into account all expected failure modes.

The FIT figures of the internal black boxes shall be combined to give the failure rate of the OFA as explained in Table B.1.

Table B.1 – Failure rate of components

Element	Number of elements	Measured value (UCL 60 %)
Doped fibre	n_1	A_1 FIT (random failure)
Splice	n_2	A_2 FIT (random failure)
Optical connector	n_3	A_3 FIT (random failure)
Electronics	n_4	A_4 FIT (random failure)
Pump laser module type 1	$n_{(4+1)}$	$A_{(4+1)}$ FIT (random and wear-out failure)
Pump laser module type 2	$n_{(4+2)}$	$A_{(4+2)}$ FIT (random and wear-out failure)
.....		
Pump laser module type m	$n_{(4+m)}$	$A_{(4+m)}$ FIT (random and wear-out failure)
Monitor diode module type 1	$n_{(4+m+1)}$	$A_{(4+m+1)}$ FIT (random failure)
Monitor diode module type 2	$n_{(4+m+2)}$	$A_{(4+m+2)}$ FIT (random failure)
Monitor diode module type h	$n_{(4+m+h)}$	$A_{(4+m+h)}$ FIT (random failure)
Passive optical component type 1	$n_{(4+m+h+1)}$	$A_{(4+m+h+1)}$ FIT (random failure)
Passive optical component type 2	$n_{(4+m+h+2)}$	$A_{(4+m+h+2)}$ FIT (random failure)
.....		
Passive optical component type k	$n_{(4+m+h+k)}$	$A_{(4+m+h+k)}$ FIT (wear-out failure)
Variable optical attenuator type 1	$n_{(4+m+h+k+1)}$	$A_{(4+m+h+k+1)}$ FIT (random failure)
Variable optical attenuator type 2	$n_{(4+m+h+k+2)}$	$A_{(4+m+h+k+2)}$ FIT (random failure)
.....		
Variable optical attenuator type l	$n_{(4+m+h+k+l)}$	$A_{(4+m+h+k+l)}$ FIT (random failure)
Total optical amplifier		$\sum_i A_i * n_i$
NOTE 1 n_i is the number of components of each type included in the OFA.		
NOTE 2 If there is some redundancy in the pump source configuration, the FIT rate to be declared is that of the pump module as a whole (i.e. taking into account that a secondary laser is turned on when the primary laser fails).		
NOTE 3 60 % UCL is common; however, 90 % can be used if deemed necessary.		

The wear-out failure rate for the pump laser module is calculated according to the real end-of-life criteria of the optical amplifier, which may be more stringent than the current reliability model for other laser modules. The random failure rate will be inferred for other components from tests performed for reliability qualification or from field data (that is, splices, optical connectors).

B.2 Guidance on failure rate calculations

Refer to IEC 62005-2.

Bibliography

IEC 60068-2-1, *Environmental testing – Part 2-1: Tests – Test A: Cold*

IEC 60068-2-6, *Environmental testing – Part 2-6: Tests – Test Fc: Vibration (sinusoidal)*

IEC 61300-2-1, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 2-1: Tests – Vibration (sinusoidal)*

IEC 61300-2-2, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 2-2: Tests – Mating durability*

IEC 61300-2-5, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 2-5: Tests – Torsion*

IEC 61300-2-9, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 2-9: Tests – Shock*

IEC 61300-2-18, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 2-18: Tests – Dry heat – High temperature endurance*

IEC 61300-2-19, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 2-19: Tests – Damp heat (steady state)*

IEC 61300-2-22, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 2-22: Tests – Change of temperature*

IEC 61300-2-42, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 2-42: Tests – Static side load for strain relief*

IEC 61300-2-44, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 2-44: Tests – Flexing of the strain relief of fibre optic devices*

IEC 62005-2, *Reliability of fibre optic interconnecting devices and passive components – Part 2: Quantitative assessment of reliability based on accelerated ageing test – Temperature and humidity; steady state*

IEC 62343-2, *Dynamic modules – Part 2: Reliability qualification*

IEC TR 62572-2:2008, *Fibre optic active components and devices – Reliability standards – Part 2: Laser module degradation*

IEC 62572-3, *Fibre optic active components and devices – Reliability standards – Part 3: Laser modules used for telecommunication*

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ITU-T G.661, *Definition and test methods for the relevant generic parameters of optical amplifier devices and subsystems*

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